

An Exact Obstruction to Joint Sequential Continuity

Abstract measure topology on L-embedded Banach spaces

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Abstract

Peralta and Pfitzner ask whether addition is sequentially jointly continuous for the abstract measure topology on every L-embedded Banach space. We give an exact obstruction reduction. If addition fails, two seminormalized asymptotically isometric copies of the unit vector basis of ℓ_1 can be chosen whose termwise sum is seminormalized and either (i) is an ℓ_1 -basic sequence with no almost isometric ℓ_1 subsequence, exhibiting strict norm loss at every singular weak-star cluster point, or (ii) is weakly null. Conversely, either configuration forces failure. In the weakly null branch the two summands have the same limiting norm and opposite singular weak-star limits. A simple non-L-embedded max-sum realizes the second branch, showing that the remaining issue is genuinely tied to L-embedded geometry. The universal question is not resolved here.

Packet status

partial_result_likely_valid. The reduction and its converse are proved. This packet does not claim that the two obstructions are impossible, or that either occurs, in an L-embedded Banach space.

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1 Source question

Let X be L-embedded, with its canonical L-decomposition

$$X^{**} = X \oplus_1 X_s.$$

Pfitzner defined the abstract measure topology τ_μ by prescribing its convergent sequences [2]. Peralta and Pfitzner recall the definition and then ask the following on page 6 of arXiv:1505.05302 [1].

By definition addition is separately continuous with respect to τ_μ , but it is an open problem whether addition is sequentially jointly continuous or even jointly continuous on an L-embedded space X .

In [24, §5] a topology, called the *abstract measure topology*, has been defined on L-embedded Banach spaces X , with the particularity that if X is a commutative or a non-commutative L^1 with finite measure/trace, this topology coincides on (norm) bounded sets with the usual measure topology. The abstract measure topology of an L-embedded space X will be denoted by τ_μ . We recall some properties of τ_μ . It is a sequential topology, which means that sets are closed if they are sequentially closed. Therefore τ_μ is determined by its convergent sequences: a sequence (x_n) converges to x in τ_μ if and only if $(x_n - x)$ is τ_μ -null and a sequence (y_n) is τ_μ -null if and only if each subsequence (y_{n_k}) contains a further subsequence $(y_{n_{k_m}})$ which either is norm null or is semi-normalized (i.e. bounded and $\inf \|y_{n_{k_m}}\| > 0$) and such that $(y_{n_{k_m}} / \|y_{n_{k_m}}\|)$ spans ℓ_1 almost isometrically. In particular, τ_μ -convergent sequences are (norm) bounded. (Note that in [24] the definition of 'to span ℓ_1 almost isometrically' differs slightly from ours but coincides with ours for normalized sequences.) It is not known whether τ_μ is Hausdorff but convergent sequences have unique limits (cf. [24, Proof of Theorem 5.2]). By definition addition is separately continuous with respect to τ_μ , but it is an open problem whether addition is sequentially jointly continuous or even jointly continuous on an L-embedded space X (cf. [24, Question 2]). It can be shown from the results in [27], that addition is τ_μ -sequentially continuous on the unit ball of a von Neumann algebra predual (see [24, Comments in Question 2]). In this section we shall enlarge the examples list by showing that a similar statement remains true for JBW*-triple preduals.

Bounded sets of an L-embedded space satisfy a mild form of sequential compactness

Figure 1: The source question and the sequential description of τ_μ , page 6 of arXiv:1505.05302.

The same paper proves sequential joint continuity when X is a JBW*-triple predual. The question for arbitrary L-embedded spaces remains the target of this packet.

2 Definitions and preliminary facts

Definition 2.1. A sequence (u_n) of nonzero vectors *spans ℓ_1 asymptotically isometrically* if there are $\delta_n \in [0, 1)$ with $\delta_n \rightarrow 0$ such that, for every finitely supported scalar sequence (a_n) ,

$$\sum_n (1 - \delta_n) |a_n| \leq \left\| \sum_n a_n \frac{u_n}{\|u_n\|} \right\| \leq \sum_n |a_n|.$$

It is *seminormalized* if $0 < \inf_n \|u_n\| \leq \sup_n \|u_n\| < \infty$.

For an L -embedded space, a sequence is τ_μ -null exactly when it is bounded and every subsequence has a further subsequence which is norm null or whose normalization spans ℓ_1 almost isometrically [2, 1]. Pfitzner's perturbation result allows "almost isometrically" to be replaced, after one more subsequence, by "asymptotically isometrically." We use that equivalent sequential description below.

We also use three standard facts.

- (i) Every weak-star accumulation point of an asymptotically isometric ℓ_1 sequence in an L -embedded space lies in X_s , by Pfitzner's fundamental weak-star accumulation lemma [2].
- (ii) A bounded sequence has either an ℓ_1 -basic subsequence or a weakly Cauchy subsequence, by Rosenthal's ℓ_1 theorem [3].
- (iii) L -embedded Banach spaces are weakly sequentially complete [4, Theorem IV.2.2].

The following norm refinement of (i) is elementary but useful.

Lemma 2.2 (No norm loss for an asymptotic summand). *Let (u_n) span ℓ_1 asymptotically isometrically in an L -embedded space, and suppose $\|u_n\| \rightarrow a > 0$. Every weak-star accumulation point u^{**} of (u_n) belongs to X_s and satisfies $\|u^{**}\| = a$.*

Proof. Membership in X_s is Pfitzner's result recalled above. Put $q_n = u_n/\|u_n\|$. If a subnet of (u_n) converges weak-star to u^{**} , the corresponding subnet of (q_n) converges to $q^{**} = u^{**}/a$.

Fix $\varepsilon > 0$ and choose N so that $\delta_n \leq \varepsilon$ for $n \geq N$. On the closed span of $(q_n)_{n \geq N}$, the formula

$$f_N \left(\sum_{n \geq N} c_n q_n \right) = \sum_{n \geq N} c_n$$

defines a functional of norm at most $(1 - \varepsilon)^{-1}$. Extend it to X by Hahn–Banach. Along any free subnet of the tail, $f_N(q_n) = 1$, and hence $f_N(q^{**}) = 1$. Thus $\|q^{**}\| \geq 1 - \varepsilon$. Since $\|q^{**}\| \leq 1$ and ε is arbitrary, $\|q^{**}\| = 1$. $\square$

3 Proof intuition

Failure of joint sequential continuity means that two τ_μ -null sequences have a sum which fails the subsequence test defining τ_μ . After discarding the easy norm-null cases, both summands must be seminormalized asymptotically isometric ℓ_1 sequences. Their weak-star cluster points therefore live, with full norm, in the singular summand X_s . Consequently every weak-star cluster point of their termwise sum also lies in X_s .

Rosenthal's dichotomy now leaves only two possibilities for a bad sum. It either retains an ℓ_1 -basic subsequence but never improves to an almost isometric one; this is the singular norm-loss branch. Or it has no ℓ_1 subsequence, hence has a weakly convergent subsequence. Its weak limit belongs simultaneously to X and X_s , so it must be zero. This is the weak-null cancellation branch. Reversing the argument is immediate: neither branch can pass the defining τ_μ subsequence test.

4 The exact obstruction theorem

Theorem 4.1 (Two-branch obstruction). *Let X be an L -embedded Banach space with abstract measure topology τ_μ . Addition fails to be sequentially jointly continuous if and only if there are seminormalized sequences (x_n) and (y_n) , each spanning ℓ_1 asymptotically isometrically, such that $z_n = x_n + y_n$ is seminormalized and, after passing to a subsequence, one of the following mutually exclusive alternatives holds.*

- (A) **Distorted- ℓ_1 / singular norm-loss branch.** (z_n) is equivalent to the unit vector basis of ℓ_1 , but no subsequence of $(z_n/\|z_n\|)$ spans ℓ_1 almost isometrically. Equivalently via Pfitzner's cluster criterion, every weak-star accumulation point of $(z_n/\|z_n\|)$ lies in X_s and has norm strictly less than one.
- (B) **Weak-null cancellation branch.** (z_n) is weakly null. Moreover, the sequences may be chosen so that $\|x_n\| \rightarrow a$, $\|y_n\| \rightarrow a$ for some $a > 0$, and every common weak-star cluster pair (x_s, y_s) satisfies $x_s, y_s \in X_s$ and $y_s = -x_s$.

Proof. Suppose first that addition is not sequentially jointly continuous. There are τ_μ -null sequences (a_n) and (b_n) such that $c_n = a_n + b_n$ is not τ_μ -null. By the defining subsequence criterion, there is a subsequence of (c_n) with no further subsequence which is norm null or whose normalization spans ℓ_1 almost isometrically. Relabel this subsequence as (c_n) .

Apply the τ_μ -null criterion first to (a_n) and then, inside the resulting subsequence, to (b_n) . A norm-null alternative for either summand is impossible: if the other summand is norm null then so is (c_n) ; if the other is an almost/asymptotically isometric ℓ_1 sequence, adding a norm-null sequence preserves that property after separating the norms away from zero. Either conclusion contradicts the choice of (c_n) . Passing to further subsequences and using Pfitzner's almost-to-asymptotic extraction, we therefore obtain seminormalized asymptotically isometric ℓ_1 sequences (x_n) and (y_n) . We may also arrange

$$\|x_n\| \rightarrow a > 0, \quad \|y_n\| \rightarrow b > 0,$$

and $z_n = x_n + y_n$ remains seminormalized and has no good subsequence.

Every weak-star accumulation point of (z_n) lies in X_s . Indeed, given a weak-star convergent subnet of (z_n) , pass to a further subnet on which the bounded pair (x_n, y_n) converges weak-star to (x_s, y_s) . By [theorem 2.2](#), $x_s, y_s \in X_s$, and hence the limit $x_s + y_s$ of the sum belongs to X_s .

If (z_n) has an ℓ_1 -basic subsequence, pass to it. It still has no almost isometric ℓ_1 subsequence. Pfitzner's cluster criterion says that a norm-one weak-star accumulation point in X_s would produce an asymptotically, hence almost, isometric ℓ_1 subsequence. Thus every normalized cluster point has norm less than one, and alternative (A) holds.

Suppose instead that (z_n) has no ℓ_1 -basic subsequence. Rosenthal's theorem gives a weakly Cauchy subsequence. Since X is weakly sequentially complete, pass to a weakly convergent subsequence, say $z_n \rightharpoonup z$ with $z \in X$. The same z , viewed in X^{**} , is a weak-star accumulation point of (z_n) , so the preceding paragraph gives $z \in X_s$. The L-decomposition is direct, hence $z \in X \cap X_s = \{0\}$. Thus (z_n) is weakly null and alternative (B) holds.

For the final assertion in (B), take a common weak-star cluster pair (x_s, y_s) . Weak convergence of the sum gives $x_s + y_s = 0$. By [theorem 2.2](#), $\|x_s\| = a$ and $\|y_s\| = b$, so $a = b$ and $y_s = -x_s$.

Conversely, assume either (A) or (B). Each of (x_n) and (y_n) is τ_μ -null. In (A), (z_n) is seminormalized and has no almost isometric ℓ_1 subsequence, so it is not τ_μ -null. In (B), a weakly null sequence cannot have an ℓ_1 -basic subsequence: on the span of such a subsequence the functional taking value one on every basis vector is bounded and extends to X , contradicting weak nullity. Since (z_n) is also seminormalized, it has neither kind of good subsequence and is not τ_μ -null. Addition therefore fails to be sequentially jointly continuous. $\square$

Corollary 4.2 (Operator form of branch (B)). *Branch (B) occurs if and only if there are asymptotically isometric embeddings $U, V : \ell_1 \rightarrow X$ such that*

$$(U + V)e_n \rightharpoonup 0, \quad \inf_n \|(U + V)e_n\| > 0,$$

after harmless positive rescaling of the two bases by factors converging to the same constant.

Proof. In branch (B), write $\|x_n\| \rightarrow a$ and $\|y_n\| \rightarrow a$, and set $q_n = x_n/\|x_n\|$, $r_n = y_n/\|y_n\|$. Then

$$q_n + r_n = \frac{z_n}{a} + \left(\frac{1}{\|x_n\|} - \frac{1}{a} \right) x_n + \left(\frac{1}{\|y_n\|} - \frac{1}{a} \right) y_n.$$

The last two terms tend to zero in norm. Hence $(q_n + r_n)$ is weakly null and, after deleting finitely many terms, seminormalized. Define $Ue_n = q_n$ and $Ve_n = r_n$. The asymptotic lower estimates make U and V embeddings. The converse follows directly from the unit-vector images. $\square$

This operator formulation may be useful because the operator $U + V$ is weakly compact: the weakly null set $\{(U + V)e_n : n \in \mathbb{N}\} \cup \{0\}$ is weakly compact, and its closed absolutely convex hull contains the image of the unit ball of ℓ_1 .

5 A sharpness model outside the L-embedded class

The weak-null obstruction is geometrically feasible without L-embeddedness. Let

$$E = \ell_1 \oplus_\infty c_0$$

and write (e_n) for the unit vector basis in either coordinate. Define

$$x_n = (e_n, 0), \quad y_n = (-e_n, e_n), \quad z_n = x_n + y_n = (0, e_n).$$

Both (x_n) and (y_n) span ℓ_1 isometrically. Indeed, for finitely supported (a_n) ,

$$\left\| \sum_n a_n y_n \right\| = \max \left\{ \sum_n |a_n|, \sup_n |a_n| \right\} = \sum_n |a_n|.$$

Yet (z_n) is weakly null and has norm one. Thus it is exactly the branch (B) configuration.

This is not a counterexample to the source question. The space E is not L-embedded: L-embedded spaces are weakly sequentially complete, while the c_0 summand contains the weakly Cauchy sequence $e_1 + \cdots + e_n$, which has no weak limit in c_0 .

6 Verification, limitations, and novelty

No computation is used. The proof should be reviewed at three points:

- (1) the double subsequence extraction from the definition of τ_μ ;
- (2) the passage from individual singular cluster points to singular cluster points of the termwise sum;
- (3) the use of Rosenthal's theorem and weak sequential completeness to turn the non- ℓ_1 branch into a weakly null sequence.

The theorem does not exclude either obstruction in an arbitrary L-embedded space. In particular, it is not a full solution of the source question. Its contribution is an exact reduction to two concrete sequence-geometric configurations, including a complete description of the cancellation in the weak-null branch.

A bounded literature search through 19 July 2026 used the exact phrases “abstract measure topology” with “addition”, “sequentially jointly continuous”, and “weakly null”, as well as searches for termwise sums of asymptotically isometric ℓ_1 sequences. It found the source papers and later applications of abstract measure topology, but no later proof, counterexample, or this two-branch reduction. This is evidence for review, not a claim of priority or an exhaustive citation search.

7 Human-review recommendation

Review as a likely valid substantial partial reduction. The most promising next target is branch (B): determine whether an L -embedded space can contain two asymptotically isometric ℓ_1 embeddings whose sum is weakly compact but remains seminormalized on the unit-vector basis. Branch (A) is the secondary target and asks whether the singular cluster points of such a sum can all lose norm.

References

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